

Organization Data Filtering and Analysis using SQL

Introduction

In this project, I used SQL queries to extract and analyse information from an organization's database within a virtual machine (VM) environment. The goal was to practice retrieving and filtering specific data using the WHERE and LIKE operators.

The project involved exploring multiple tables within the database, including employee and device information. I ran SQL queries to identify details such as device IDs, operating systems, and operating system versions. Additionally, I filtered employee data to determine how many employees work in specific departments and to identify which machines or devices belong to employees I particular office locations.

This project demonstrates practical use of SQL for filtering, pattern matching, and analysing organizational data in s secure, simulated environment.

Task 1: List all Organization machines and their operating system

Objective:

To retrieve and display a complete list of all devices within the organization along with their operating system information.

```
MariaDB [organization]> select device_id, operating_system  
-> from machines;
```

device_id	operating_system
a184b775c707	OS 1
a192b174c940	OS 2
a305b818c708	OS 3
a317b635c465	OS 1
a320b137c219	OS 2
a398b471c573	OS 3
a667b270c984	OS 1
a821b452c176	OS 2

Command: Select device_id, operating_system
from machines;

Explanation: This query retrieves the device_id and operating_system columns from the machines table, effectively listing all devices in the organization along with their operating system.

Task 2: Identify How Many Devices Are Running OS 2 For Emergency Update

Objective: To filter the devices in the organization's database and identify those running OS 2, which requires an urgent update. This task demonstrates the use of the WHERE clause to retrieve specific records based on condition, supporting proactive device management and security maintenance.

MariaDB [organization]> select device_id, operating_system -> from machines -> where operating_system = 'OS 2';		y347z204a710	OS 2
		y765z123a548	OS 2
		y976z753a267	OS 2
		z451a308b518	OS 2
		z654a154b259	OS 2
		80 rows in set (0.001 sec)	

Command: Select device_id, operating_system
From machines
Where operating_system = 'OS 2'.

Explanation: From the results, we find that there are 80 devices running OS 2, highlighting the scope of the emergency update needed. This targeted approach is essential for efficient maintenance and reading potential security risks associated with outdated operating systems.

Task 3 Retrieve Employees in Finance and Their Office Numbers

Objective: To filter the employee records in the organization’s database and identify all employees working in the Finance and along with their office numbers.

MariaDB [organization]> Select * -> from employees -> where department = 'finance';					
employee_id	device_id	username	department	office	
1003	d394e816f943	sgilmore	Finance	South-153	
1007	h174i497j413	wjaffrey	Finance	North-406	
1008	i858j583k571	abernard	Finance	South-170	
1010	k242l212m542	jlansky	Finance	South-109	
1015	p611q262r945	jsoto	Finance	North-271	
1017	r550s824t230	jclark	Finance	North-188	

Command: Select*
From employees
Where department = 'finance';

Explanation: By using the WHERE clause, it filters the results to include only employees who work in the Finance department. We can see the employees working in the finance department and their office numbers

Task 3: Identify Employee Using the Unresponsive Machine in the South-109 Building

Objective: To determine which employee is assigned to the machine located in the South-109 building that has reported an issue.

```
MariaDB [organization]> select *
-> from employees
-> where office = 'south-109';
```

employee_id	device_id	username	department	office
1010	k2421212m542	jlansky	Finance	South-109

Command: select *
From employees
Where office = 'south-109';

Explanation: This task involved combining data from the machines and employees table to identify which employee was linked to the unresponsive system in the South-109 building

Findings:

The query results showed that the computer in office South-109 is assigned to **employee ID 1010**, With the user **jlansky**, who works in the **Finance Department**.

Task 4: Identify All Machines Located in the South Building

Objective: To identify all machines located in the South building, as the IT team has determined there are issues affecting all systems in that location

```
MariaDB [organization]> select *
-> from employees
-> where office like 'south%';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k2421212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1013	n205o559p243	zbernal	Information Technology	South-229
1020	u899v381w363	arutley	Marketing	South-351
1021	v200w121x977	smartell	Information Technology	South-138

Command: select *
From employees
Where office like 'south%';

Explanation: To locate the only machines in the South building, a SQL query was used with the LIKE operator and the % wildcard – for example, matching office names that begin with “south”. The % symbol acts as a placeholder for any sequence of characters that follows, allowing the query to find every record where the office name starts with “south”, regardless of the specific room number.

The query allowed all the identified machines were isolated for troubleshooting and detailed inspection by the IT team

Conclusion/What I Learned

In this project, I learned how to use SQL to retrieve and filter data from an organization's database to investigate and solve real- world IT issues. The project focused on using key filtering commands to refine query results and identify affected systems efficiently.

Key SQL commands Learned

- WHERE clause – Used to filter results based on specific conditions, such as finding employees in certain departments or machines located in specific buildings.
- LIKE operator – Used to search for patterns in data, for example, locating all offices beginning with "South" to identify affected devices.

Practical Outcomes

- Identified devices matching specific conditions (e.g., building or OS)
- Determined which employees were using affected computers.
- Isolated all machines located in the South building for troubleshooting and inspection.

This project enhanced my understanding of how SQL can be used to query, filter, and analyse organizational data effectively for IT system management and issue resolution.